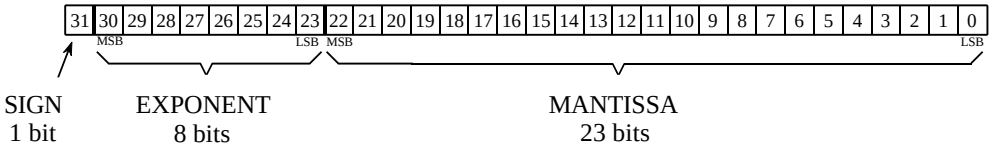
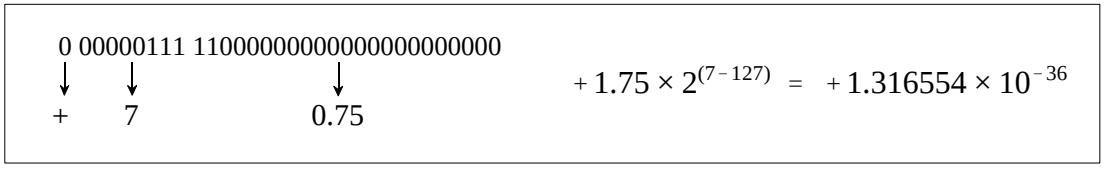


Example 1



Example 2

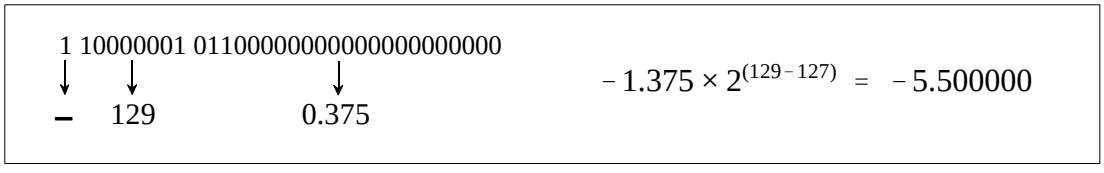


FIGURE 4-2
Single precision floating point storage format. The 32 bits are broken into three separate parts, the sign bit, the exponent and the mantissa. Equations 4-1 and 4-2 shows how the represented number is found from these three parts. MSB and LSB refer to "most significant bit" and "least significant bit," respectively.

base 2). Since the only nonzero number that exists in base two is 1, the leading digit in the mantissa will always be a 1, and therefore does not need to be stored. Removing this redundancy allows the number to have an additional one bit of precision. The 23 stored bits, referred to by the notation: $m_{22}, m_{21}, m_{21}, \cdots, m_0$, form the mantissa according to:

EQUATION 4-2
Algorithm for converting the bit pattern into the mantissa, *M*, used in Eq. 4-1.

$$M = 1.m_{22}m_{21}m_{20}m_{19}\cdots m_2m_1m_0$$

In other words, $M = 1 + m_{22}2^{-1} + m_{21}2^{-2} + m_{20}2^{-3}\cdots$. If bits 0 through 22 are all *zeros*, *M* takes on the value of one. If bits 0 through 22 are all *ones*, *M* is just a hair under two, i.e., $2 - 2^{-23}$.

Using this encoding scheme, the largest number that can be represented is: $\pm(2-2^{-23}) \times 2^{128} = \pm 6.8 \times 10^{38}$. Likewise, the smallest number that can be represented is: $\pm 1.0 \times 2^{-127} = \pm 5.9 \times 10^{-39}$. The IEEE standard reduces this range slightly to free bit patterns that are assigned special meanings. In particular, the largest and smallest numbers allowed in the standard are